

Constrained Committee Selection with Candidate-Level Approval Ballots and Attribute Quota Constraints

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Abstract

We study a novel variant of approval-based committee selection where voters approve candidates directly, each candidate possesses d quantifiable attributes, and the selected committee must satisfy externally imposed *marginal quota constraints* on its attribute composition. Unlike prior work on constrained multiwinner voting—which models constraints as group fairness or diversity conditions—our constraints are explicit per-attribute threshold requirements (e.g., “at least 4 members must have rating ≥ 4 , and at least 3 must have age > 35 ”). We formalize this problem, establish its hardness, and propose CIGAV (Constraint-Integrated Greedy Approval Voting), a polynomial-time algorithm that adaptively balances voter representation against constraint satisfaction via a dynamic weighting mechanism. We prove an approximation ratio of $1 + Q/k - Q$ where Q is the total quota demand and k is the committee size, showing that CIGAV approaches optimality when quota demands are small relative to committee size. CIGAV strictly reduces to Greedy Approval Voting (GAV) when no constraints are present, inheriting its Justified Representation (JR) guarantee.

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1 Introduction

Committee selection from a set of candidates by aggregating voter preferences is a fundamental problem in social choice theory [1, 2]. In many real-world settings, however, the selected committee must also satisfy domain-specific requirements on its attribute composition. For example, a hiring committee may require at least 4 members with a performance rating above 4, at least 3 with intermediate-or-higher education, and at least 3 above age 35—all simultaneously.

Motivation. Standard approval-based committee (ABC) voting rules such as Approval Voting (AV), Proportional Approval Voting (PAV), and Greedy Approval Voting (GAV) [1, 4] optimize for voter representation but are entirely agnostic to candidate attributes. When attribute constraints are imposed externally (by an employer, a regulation, or a policy), purely approval-driven rules can produce committees that violate these requirements. Conversely, methods that maximize constraint satisfaction without regard to voter preferences may produce committees that leave large voter groups entirely unrepresented.

Our Contribution. We make the following contributions:

1. We formally define the *Constrained Approval-Based Committee Selection* (CACS) problem with marginal attribute quota constraints (§3).
2. We identify a fundamental *approval–constraint tension*: high-approval candidates may not satisfy attribute constraints, and constraint-satisfying candidates may be unpopular with voters.
3. We propose CIGAV, a two-phase adaptive greedy algorithm that resolves this tension via a dynamic weighting parameter β (§4).
4. We prove a polynomial-time complexity of $O(nm + kmp + kp^3)$ and an approximation ratio of $1 + Q/k$ (§5).
5. We show that CIGAV reduces to GAV when no constraints are present, inheriting its Justified Representation guarantee (§6).

2 Related Work

Approval-Based Committee Voting. Aziz et al. [1] introduced Justified Representation (JR) as a fairness axiom for approval-based committee voting and showed that GAV and PAV satisfy it. Elkind et al. [2] systematically study properties of multiwinner voting rules. Kagita et al. [3] extend the framework to attribute-level approvals where voters approve attribute values rather than candidates directly; our setting differs in that voters approve candidates directly while constraints are placed on the committee’s attribute profile.

Constrained Multiwinner Voting. Vishnoi et al. [5] introduce an algorithmic framework for multiwinner voting with fairness constraints, covering monotone and submodular

score functions. Their constraints model group fairness (e.g., gender balance) as proportional requirements. Our constraints differ: they are externally imposed *threshold quota* requirements on specific attribute values, not derived from voter demographics.

Aziz [6] proposes a cubic-time algorithm for committee elections with soft diversity constraints. Bredereck et al. [7] study multiwinner elections with diversity constraints in the AAAI setting. The DiRe Committee framework [8] simultaneously handles diversity and representation constraints but treats them as binary conditions rather than explicit quota thresholds.

Attribute-Constrained Selection. A recent paper [9] defines CECAC (Committee Election with Candidate Attribute Constraints) where candidates have a fixed profit and the goal is to maximize total profit subject to propositional constraints on attributes. CECAC does not model voters at all; our work extends this by replacing fixed profit with voter-driven approval scores, introducing the approval-constraint tension that is the central novelty of our setting.

3 Problem Formalization

3.1 Basic Setup

Let $V = \{v_1, v_2, \dots, v_n\}$ be the set of *voters* and $C = \{c_1, c_2, \dots, c_m\}$ the set of *candidates*. Each candidate c_i is associated with a d -dimensional attribute vector:

$$c_i = [a_i^1, a_i^2, \dots, a_i^d],$$

where $a_i^j \in D^j$ and D^j is the domain of attribute j .

Each voter v_i submits an *approval ballot* $A_i \subseteq C$ indicating the candidates they approve, with the constraint $1 < |A_i| < r$ for some bound r (multi-approval, bounded ballot size).

Definition 3.1 (Approval Score). *The approval score of a committee $W \subseteq C$ with respect to voter set V is:*

$$\text{AV}(W, V) = \sum_{v_i \in V} |W \cap A_i|.$$

The individual approval score of a candidate c_i is $\text{AV}(c_i, V) = |\{v_j \in V : c_i \in A_j\}|$.

3.2 Marginal Attribute Quota Constraints

A *marginal constraint* is a triple $\gamma_t = (j, P_t, q_t)$ where:

- $j \in [1, d]$ is the attribute dimension,
- $P_t : D^j \rightarrow \{0, 1\}$ is a predicate (e.g., `rating` ≥ 4),
- $q_t \in \mathbb{Z}_{>0}$ is the minimum number of committee members that must satisfy P_t .

Let $\Gamma = \{\gamma_1, \gamma_2, \dots, \gamma_p\}$ be the constraint set.

Definition 3.2 (Constraint Satisfaction Vector). *For each candidate c_i , define the constraint satisfaction vector $\mathbf{s}_i = [s_i^1, s_i^2, \dots, s_i^p] \in \{0, 1\}^p$ where*

$$s_i^t = \mathbf{1}[c_i \text{ satisfies predicate } P_t].$$

A committee W is *feasible* if and only if

$$\sum_{c_i \in W} s_i^t \geq q_t \quad \forall t \in [1, p].$$

Example 3.3. Let the constraints be: $\gamma_1 = (\text{rating}, \geq 4, 4)$, $\gamma_2 = (\text{education}, = \text{intermediate}, 3)$, $\gamma_3 = (\text{age}, > 35, 3)$. Then $s_i^1 = 1$ iff c_i 's rating is at least 4, and so on. A candidate with rating 4.5, intermediate education, and age 40 has $\mathbf{s}_i = [1, 1, 1]$ and contributes to all three quotas.

3.3 Problem Statement

Definition 3.4 (CACS Problem). Given voters V , candidates C (with d attributes each), approval ballots $\{A_i\}_{v_i \in V}$ with $1 < |A_i| < r$, constraints $\Gamma = \{\gamma_1, \dots, \gamma_p\}$, and committee size k , the Constrained Approval-Based Committee Selection (CACS) problem asks for a committee $W \subseteq C$ with $|W| = k$ that:

1. Satisfies all constraints: $\sum_{c_i \in W} s_i^t \geq q_t$ for all $t \in [1, p]$,
2. Maximizes the approval score: $\text{AV}(W, V) = \sum_{v_i \in V} |W \cap A_i|$.

3.4 Core Tension

A fundamental conflict distinguishes CACS from unconstrained ABC voting:

Proposition 3.5 (Approval–Constraint Tension). *There exist instances where the committee maximizing $\text{AV}(W, V)$ violates at least one constraint in Γ , and every feasible committee achieves strictly lower approval score.*

Proof. Consider a single candidate c_1 approved by all n voters. Suppose c_1 satisfies no constraint in Γ , while all constraint-satisfying candidates are approved by zero voters. Then the approval-optimal committee includes c_1 but violates all constraints, while every feasible committee has $\text{AV}(W, V) = 0 < n$. $\square$

4 The CIGAV Algorithm

4.1 Key Subroutines

4.1.1 Feasibility Check

Definition 4.1 (Feasibility). A partial committee W with $|W| \leq k$ is extendable to a feasible solution if there exists a set $W' \supseteq W$ with $|W'| = k$ satisfying all constraints in Γ .

We check extendability via an LP relaxation:

Algorithm 1 FeasibilityCheck

Require: Partial committee W , candidate set C , constraints Γ , size k

Ensure: FEASIBLE or INFEASIBLE

```

1: for each  $\gamma_t = (j, P_t, q_t) \in \Gamma$  do
2:    $C_t \leftarrow \{c_i \in C \setminus W : s_i^t = 1\}$ 
3:    $\text{deficit}_t \leftarrow \max(0, q_t - \sum_{c \in W} s_c^t)$ 
4:   if  $|\text{deficit}_t| > 0$  and  $|C_t| < \text{deficit}_t$  then return INFEASIBLE
5:   end if
6: end for
7: Solve LP: find  $\mathbf{x} \in [0, 1]^{|C \setminus W|}$  s.t.  $\sum_i x_i \leq k - |W|$ ,  $\sum_i s_i^t x_i \geq q_t - \sum_{c \in W} s_c^t$  for all  $t$ 
8: if LP is infeasible then return INFEASIBLE
9: end if return FEASIBLE

```

4.1.2 Constraint Urgency

Definition 4.2 (Constraint Urgency). *Given partial committee W and $\ell = k - |W|$ remaining slots, the urgency of constraint γ_t is:*

$$\text{Urgency}(\gamma_t, W, k) = \frac{\max(0, q_t - \sum_{c \in W} s_c^t)}{k - |W|}.$$

The overall constraint pressure is:

$$\text{Pressure}(W, k) = \max_{t \in [1, p]} \text{Urgency}(\gamma_t, W, k).$$

When $\text{Pressure}(W, k) = 1$, every remaining slot must satisfy at least one specific constraint. When $\text{Pressure}(W, k) = 0$, all constraints are already satisfied.

4.1.3 LP Feasibility Check: Illustrative Example

We illustrate why the LP feasibility check is necessary for overlapping constraints and how it works without enumerating combinations.

Setup. Consider $m = 4$ candidates, $k = 2$, with two constraints: $\gamma_1 = (\text{race} = \text{black}, q_1 = 2)$ and $\gamma_2 = (\text{gender} = \text{female}, q_2 = 2)$.

Table 1: Candidate pool for feasibility example.

Candidate	Race	Gender	s^1 (black)	s^2 (female)
c_1	Black	Male	1	0
c_2	Black	Male	1	0
c_3	White	Female	0	1
c_4	White	Female	0	1

Why Greedy Check Fails. The greedy feasibility check evaluates each constraint independently:

- γ_1 : candidates $\{c_1, c_2\}$ satisfy race = black. Available count = $2 \geq q_1 = 2$. **Reports feasible.**

- γ_2 : candidates $\{c_3, c_4\}$ satisfy gender = female. Available count = $2 \geq q_2 = 2$.
Reports feasible.

The greedy check concludes FEASIBLE — but this is **incorrect**. Satisfying γ_1 requires both slots for $\{c_1, c_2\}$, leaving no slots for γ_2 , and vice versa. No committee of size $k = 2$ can satisfy both constraints simultaneously.

LP Feasibility Check. The LP does not enumerate the $\binom{4}{2} = 6$ possible committees. Instead, it introduces a continuous variable $x_i \in [0, 1]$ for each candidate representing fractional membership, and checks whether the following system of linear inequalities is simultaneously satisfiable:

$$\sum_{i=1}^4 x_i = k = 2, \quad (1)$$

$$x_1 + x_2 \geq q_1 = 2 \quad (\gamma_1), \quad (2)$$

$$x_3 + x_4 \geq q_2 = 2 \quad (\gamma_2), \quad (3)$$

$$x_i \in [0, 1] \quad \forall i. \quad (4)$$

The LP solver derives the contradiction algebraically in a single pass, without trying any combination:

1. From constraint (2): $x_1 + x_2 \geq 2$. Since $x_i \leq 1$, this forces $x_1 = x_2 = 1$.
2. Substituting into constraint (1): $1 + 1 + x_3 + x_4 = 2$, which forces $x_3 = x_4 = 0$.
3. Substituting into constraint (3): $0 + 0 = 0 \geq 2$ — **contradiction**.

The LP correctly returns INFEASIBLE.

Remark 4.3 (LP is Not Brute Force). *The LP feasibility check does not enumerate candidate combinations. It treats committee membership as continuous variables and checks satisfiability of a linear system using the Simplex or Interior Point method. For p constraints and m candidates, this runs in $O(p^3)$ time — independent of m and exponentially faster than brute force $O(\binom{m}{k})$.*

Table 2: Greedy check vs. LP check on overlapping constraints.

Method	Answer	Correct?
Greedy (independent per constraint)	FEASIBLE	No
LP (joint constraint system)	INFEASIBLE	Yes
Brute force (all $\binom{4}{2} = 6$ committees)	INFEASIBLE	Yes

Comparison of Methods. The LP achieves the correctness of brute force at a fraction of the computational cost, making it the appropriate feasibility check for overlapping constraints in .

4.2 The CIGAV Algorithm

The key insight of CIGAV is an *adaptive weighting* between voter approval and constraint urgency. Define the combined score of a candidate $c_i \notin W$ as:

$$\text{Score}(c_i, W, V') = \underbrace{\alpha \cdot \text{AV}(c_i, V')}_{\text{approval component}} + \underbrace{\beta \cdot \sum_{t=1}^p s_i^t \cdot \text{Urgency}(\gamma_t, W, k)}_{\text{constraint urgency component}},$$

where $\beta = \text{Pressure}(W, k)$ and $\alpha = 1 - \beta$. This self-regulating mechanism automatically shifts priority from voter satisfaction to constraint satisfaction as deadlines approach.

Algorithm 2 CIGAV: Constraint-Integrated Greedy Approval Voting

Require: $C, V, k, \{A_i\}_{v_i \in V}, \Gamma = \{\gamma_1, \dots, \gamma_p\}, d$ **Ensure:** $W \subseteq C, |W| = k$ satisfying Γ , or $\emptyset$ if infeasible

```
1: // Preprocessing
2: Compute  $s_i^t$  for all  $c_i \in C, t \in [1, p]$   $O(pm)$ 
3: Compute  $AV(c_i, V)$  for all  $c_i \in C$   $O(nm)$ 
4: if FEASIBILITYCHECK( $C, \emptyset, \Gamma, k$ ) = INFEASIBLE then return  $\emptyset$ 
5: end if
6:  $W \leftarrow \emptyset; V' \leftarrow V$ 
7: // Phase 1: Forced Selection
8: for each  $\gamma_t = (j, P_t, q_t) \in \Gamma$  do
9:    $C_t \leftarrow \{c_i \in C : s_i^t = 1\}$ 
10:  if  $|C_t| = q_t$  then  $\triangleright$  All constraint-satisfying candidates are mandatory
11:     $W \leftarrow W \cup C_t$ 
12:     $q_t \leftarrow 0$   $\triangleright$  Constraint fully satisfied
13:  end if
14: end for
15: // Phase 2: Adaptive Greedy Selection
16: while  $|W| < k$  do
17:    $\beta \leftarrow \text{Pressure}(W, k); \alpha \leftarrow 1 - \beta$ 
18:   for each  $c_i \in C \setminus W$  do
19:      $\text{Score}(c_i) \leftarrow \alpha \cdot AV(c_i, V') + \beta \cdot \sum_t s_i^t \cdot \text{Urgency}(\gamma_t, W, k)$ 
20:   end for
21:   Sort  $C \setminus W$  by  $\text{Score}(\cdot)$  in descending order
22:   for each  $c_i$  in sorted order do  $\triangleright$  Feasibility-safe selection
23:     if FEASIBILITYCHECK( $C, W \cup \{c_i\}, \Gamma, k$ ) = FEASIBLE then
24:        $c^* \leftarrow c_i; \text{break}$ 
25:     end if
26:   end for
27:    $W \leftarrow W \cup \{c^*\}$ 
28:    $V' \leftarrow V' \setminus \{v_i \in V' : c^* \in A_i\}$ 
29:   if  $V' = \emptyset$  then
30:      $V' \leftarrow V$   $\triangleright$  Reset: all voters represented
31:   end if
32:   for each  $\gamma_t \in \Gamma$  do
33:      $q_t \leftarrow \max(0, q_t - s_{c^*}^t)$ 
34:   end for
35: end while return  $W$ 
```

4.3 Worked Example

Setup. Consider $m = 6$ candidates, $k = 3$, $n = 4$ voters, with two constraints: $\gamma_1 = (\text{rating} \geq 4, q_1 = 2)$ and $\gamma_2 = (\text{age} > 35, q_2 = 2)$.

Table 3: Candidate attributes and voter approvals.

Candidate	Rating	Age	s^1 (rating ≥ 4)	s^2 (age >35)	Approvals
c_1	4.5	40	1	1	v_1, v_2, v_3
c_2	4.2	30	1	0	v_1, v_2
c_3	3.5	38	0	1	v_2, v_3, v_4
c_4	4.1	28	1	0	v_4
c_5	3.2	42	0	1	v_3
c_6	3.8	32	0	0	v_1, v_4

Note that constraints overlap: c_1 satisfies both γ_1 and γ_2 simultaneously.

Preprocessing. Approval scores: $(c_1) = 3$, $(c_2) = 2$, $(c_3) = 3$, $(c_4) = 1$, $(c_5) = 1$, $(c_6) = 2$. The initial LP feasibility check confirms a feasible committee of size $k = 3$ exists satisfying both constraints (e.g., $\{c_1, c_2, c_3\}$).

Phase 1: Forced Selections.

- γ_1 : candidates with rating ≥ 4 are $\{c_1, c_2, c_4\}$, count = 3 $\neq q_1 = 2$. No forced selection.
- γ_2 : candidates with age > 35 are $\{c_1, c_3, c_5\}$, count = 3 $\neq q_2 = 2$. No forced selection.

$W \leftarrow \emptyset$. Proceed to Phase 2.

Phase 2: Adaptive Greedy Selection. Iteration 1. Slots remaining = 3, deficits $(d_1, d_2) = (2, 2)$, $\mathcal{V}' = \{v_1, v_2, v_3, v_4\}$.

$$\beta = \max\left(\frac{2}{3}, \frac{2}{3}\right) = 0.67, \quad \alpha = 0.33.$$

Candidate scores:

$$\begin{aligned} \text{Score}(c_1) &= 0.33 \times 3 + 0.67 \times (1 + 1) = 0.99 + 1.34 = \mathbf{2.33}, \\ \text{Score}(c_2) &= 0.33 \times 2 + 0.67 \times (1 + 0) = 0.66 + 0.67 = 1.33, \\ \text{Score}(c_3) &= 0.33 \times 3 + 0.67 \times (0 + 1) = 0.99 + 0.67 = 1.66, \\ \text{Score}(c_4) &= 0.33 \times 1 + 0.67 \times (1 + 0) = 0.33 + 0.67 = 1.00, \\ \text{Score}(c_5) &= 0.33 \times 1 + 0.67 \times (0 + 1) = 0.33 + 0.67 = 1.00, \\ \text{Score}(c_6) &= 0.33 \times 2 + 0.67 \times (0 + 0) = 0.66 + 0.00 = 0.66. \end{aligned}$$

Top candidate: c_1 . *LP check for c_1* : slots remaining after adding $c_1 = 2$, deficits drop to $(1, 1)$. LP asks: can 2 slots cover deficit 1 for γ_1 and deficit 1 for γ_2 from $\{c_2, c_3, c_4, c_5, c_6\}$? Yes (c_2 covers γ_1 ; c_3 covers γ_2). **FEASIBLE.**

Select c_1 . $W = \{c_1\}$. $\mathcal{V}' \leftarrow \emptyset \rightarrow$ reset to $\mathcal{V}$. Deficits: $(1, 1)$.

Iteration 2. Slots remaining = 2, deficits $(1, 1)$, $\mathcal{V}' = \{v_1, v_2, v_3, v_4\}$.

$$\beta = \max\left(\frac{1}{2}, \frac{1}{2}\right) = 0.5, \quad \alpha = 0.5.$$

Candidate scores (excluding c_1):

$$\begin{aligned}\text{Score}(c_2) &= 0.5 \times 2 + 0.5 \times 1 = \mathbf{1.5}, \\ \text{Score}(c_3) &= 0.5 \times 3 + 0.5 \times 1 = \mathbf{2.0}, \\ \text{Score}(c_4) &= 0.5 \times 1 + 0.5 \times 1 = 1.0, \\ \text{Score}(c_5) &= 0.5 \times 1 + 0.5 \times 1 = 1.0, \\ \text{Score}(c_6) &= 0.5 \times 2 + 0.5 \times 0 = 1.0.\end{aligned}$$

Top candidate: c_3 . *LP check for c_3* : slots remaining after = 1, deficits drop to (1, 0). LP asks: can 1 slot cover deficit 1 for γ_1 from $\{c_2, c_4, c_5, c_6\}$? Yes (c_2 or c_4 both have $s^1 = 1$). **FEASIBLE.**

Select c_3 . $W = \{c_1, c_3\}$. $\mathcal{V}' = \{v_1, v_4\}$ (unrepresented). Deficits: (1, 0).

Iteration 3. Slots remaining = 1, deficits (1, 0), $\mathcal{V}' = \{v_1, v_4\}$.

$$\beta = \max\left(\frac{1}{1}, 0\right) = 1.0, \quad \alpha = 0.0.$$

Candidate scores (excluding c_1, c_3):

$$\begin{aligned}\text{Score}(c_2) &= 0.0 \times 2 + 1.0 \times 1 = \mathbf{1.0}, \\ \text{Score}(c_4) &= 0.0 \times 1 + 1.0 \times 1 = \mathbf{1.0}, \\ \text{Score}(c_5) &= 0.0 \times 1 + 1.0 \times 0 = 0.0, \\ \text{Score}(c_6) &= 0.0 \times 2 + 1.0 \times 0 = 0.0.\end{aligned}$$

Tie between c_2 and c_4 . Break by approval in $\mathcal{V}' = \{v_1, v_4\}$: $(c_2, \mathcal{V}') = 1$ (v_1 approves c_2), $(c_4, \mathcal{V}') = 1$ (v_4 approves c_4). Still tied; break by index: select c_2 .

LP check for c_2 : slots remaining after = 0, deficits drop to (0, 0). All quotas met. **FEASIBLE.**

Select c_2 . $W = \{c_1, c_3, c_2\}$.

Final Committee. $W = \{c_1, c_2, c_3\}$ with $|W| = k = 3$.

Table 4: Constraint verification for final committee W .

Constraint	Required	Satisfied by	Status
rating ≥ 4	2	c_1, c_2	✓
age > 35	2	c_1, c_3	✓
Committee size	3	c_1, c_2, c_3	✓

Total approval: $|W \cap A_{v_1}| + |W \cap A_{v_2}| + |W \cap A_{v_3}| + |W \cap A_{v_4}| = 2 + 2 + 2 + 1 = 7$. The example demonstrates overlapping constraint satisfaction (c_1 contributes to both γ_1 and γ_2), the LP feasibility check preventing infeasible selections, and the adaptive shift of β from $0.67 \rightarrow 0.5 \rightarrow 1.0$ as constraint urgency increases.

5 Analysis

5.1 Complexity

Theorem 5.1 (Time Complexity of CIGAV). CIGAV runs in time $O(nm + kmp + kp^3)$.

Proof. We account for each phase:

- **Preprocessing:** Computing $AV(c_i, V)$ for all candidates requires scanning n ballots for each of m candidates: $O(nm)$. Computing constraint satisfaction vectors takes $O(pm)$.
- **Phase 1 (forced selection):** For each of p constraints, we scan m candidates: $O(pm)$.
- **Phase 2 (greedy loop):** The loop runs k iterations. In each iteration:
 - Score computation: $O(mp)$ per iteration.
 - Sorting: $O(m \log m)$.
 - FeasibilityCheck (LP): $O(p^3)$ per call (standard LP for p variables/constraints).
 - Total per iteration: $O(mp + p^3)$.

Over k iterations: $O(k(mp + p^3))$.

Summing: $O(nm + pm + k(mp + p^3)) = O(nm + kmp + kp^3)$. Since p is typically small in practice (e.g., $p \leq 20$), this simplifies to $O(nm + kmp)$, which is fully polynomial in all input parameters. $\square$

5.2 Approximation Ratio

Let W^* denote the optimal feasible committee with $\text{OPT} = AV(W^*, V)$, and let W_{CIGAV} denote the committee returned by CIGAV with $\text{ALG} = AV(W_{\text{CIGAV}}, V)$. Let $Q = \sum_{t=1}^p q_t$ denote the total quota demand.

Theorem 5.2 (Approximation Ratio). *Under the assumption that each voter approves at most r candidates ($|A_i| \leq r$ for all v_i), CIGAV satisfies:*

$$\frac{\text{OPT}}{\text{ALG}} \leq 1 + \frac{Q}{k - Q}.$$

Proof. We bound the approval score lost due to constraint-driven selection.

There are at most $Q = \sum_t q_t$ selection steps in which $\beta > 0$, i.e., steps where constraint pressure forces the algorithm to potentially choose a candidate with sub-optimal approval score. In each such step, the algorithm may select a candidate with approval score as low as 0 instead of a candidate with score up to n (where n is the number of voters). However, by the feasibility-safe selection in Algorithm 2, the selected candidate always maintains extendability, and at most one candidate worth up to r approvals is “missed” per constrained step (since each voter approves at most r candidates total).

Thus the total approval lost over all Q constrained steps is:

$$\text{OPT} - \text{ALG} \leq Q \cdot r.$$

Since the best possible approval score satisfies $\text{OPT} \leq k \cdot n$ (at most n approvals per committee member), and in the worst case $\text{OPT} = k \cdot r$ (every voter approves exactly one member), we have:

$$\frac{\text{OPT}}{\text{ALG}} = \frac{\text{OPT}}{\text{OPT} - (\text{OPT} - \text{ALG})} \leq \frac{k \cdot r}{k \cdot r - Q \cdot r} = \frac{k}{k - Q} = 1 + \frac{Q}{k - Q}.$$

For the practically relevant regime where $Q \leq k/2$, this simplifies to $1 + Q/k$ (up to lower-order terms). More precisely:

$$\frac{\text{OPT}}{\text{ALG}} \leq 1 + \frac{Q}{k}.$$

□

Remark 5.3. *The bound in Theorem 5.2 has the following implications:*

- **Small constraint regime** ($Q \ll k$): *The ratio approaches 1, meaning CIGAV is near-optimal.*
- **Fully constrained regime** ($Q \rightarrow k$): *The ratio grows, but in this case the committee is nearly fully determined by constraints, leaving little room for any algorithm to improve on approval score.*
- **Optimal case:** *CIGAV achieves optimal approval when constraint-satisfying candidates coincide with high-approval candidates, i.e., when FEASIBILITYCHECK never rejects the top-scored candidate.*

Corollary 5.4. *When $p = 0$ (no constraints), $Q = 0$ and the approximation ratio equals 1. CIGAV reduces exactly to GAV and achieves its approval score.*

5.3 Tightness of the Bound

The approximation bound $1 + Q/k$ is tight in the following sense:

Proposition 5.5 (Tight Instance). *There exists an instance where CIGAV achieves $\text{OPT}/\text{ALG} = 1 + Q/k$.*

Proof. Construct $m = k + Q$ candidates. Let k candidates (say $c_1, \dots, c_k$) be approved by all n voters but satisfy no constraint. Let the remaining Q candidates each satisfy exactly one constraint and be approved by 0 voters. The optimal feasible committee must include all Q constraint candidates (since they are the only ones satisfying their respective constraints) plus $k - Q$ of the popular candidates. Thus:

$$\text{OPT} = n \cdot (k - Q), \quad \text{ALG} = n \cdot (k - Q) - \epsilon$$

for any greedy that might differ in the remaining $k - Q$ steps. In the extreme case, if all Q constrained steps force zero-approval selections while the unconstrained optimum would give $Q \cdot r$ approvals:

$$\frac{\text{OPT}}{\text{ALG}} = \frac{k \cdot r}{(k - Q) \cdot r} = \frac{k}{k - Q} \approx 1 + \frac{Q}{k}.$$

□

6 Properties of CIGAV

6.1 Reduction to GAV

Theorem 6.1 (Reduction to GAV). *When $\Gamma = \emptyset$ (no constraints), CIGAV with Phase 2 operating at $\beta = 0$ throughout reduces exactly to Greedy Approval Voting (GAV).*

Proof. With $\Gamma = \emptyset$, $\text{Pressure}(W, k) = 0$ for all W , so $\beta = 0$ and $\alpha = 1$ throughout. The combined score reduces to:

$$\text{Score}(c_i) = \text{AV}(c_i, V').$$

The algorithm greedily selects the candidate with highest approval score among unrepresented voters and removes those voters upon selection—exactly the procedure of GAV [1]. $\square$

Corollary 6.2 (Inherited JR). *When $\Gamma = \emptyset$, CIGAV satisfies Simple Justified Representation (SJR), inheriting the guarantee from GAV.*

6.2 Summary of Properties

Table 5: Properties of CIGAV compared to standard rules. $\checkmark$ = satisfies, $\times$ = does not satisfy, $*$ = satisfies under no-constraint conditions ($\Gamma = \emptyset$).

Property	AV	SAV	PAV	RAV	GAV	CIGAV
Homogeneity	$\checkmark$	$\checkmark$	$\checkmark$	$\checkmark$	$\checkmark$	$\checkmark$
Consistency	$\checkmark$	$\checkmark$	$\checkmark$	$\checkmark$	$\checkmark$	$\checkmark$
Monotonicity	$\checkmark$	$\checkmark$	$\checkmark$	$\checkmark$	$\checkmark$	$\checkmark$
Constraint Feasibility	$\times$	$\times$	$\times$	$\times$	$\times$	$\checkmark$
Weak Unanimity	$\times$	$\times$	$\times$	$\times$	$\checkmark$	$\checkmark^*$
SJR	$\times$	$\times$	$\times$	$\times$	$\checkmark$	$\checkmark^*$
Polynomial Time	$\checkmark$	$\checkmark$	$\times$	$\checkmark$	$\checkmark$	$\checkmark$
Approximation Guarantee	—	—	—	—	—	$1 + Q/k$

7 Complexity of CACS

Theorem 7.1 (NP-hardness of CACS). *The decision version of the CACS problem—does there exist a feasible committee W with $\text{AV}(W, V) \geq \tau$?—is NP-complete.*

Proof Sketch. Membership in NP is immediate: given a committee W , feasibility and approval score can be verified in polynomial time.

For NP-hardness, we reduce from the Set Cover problem. Given a Set Cover instance with n' elements and subsets $S_1, \dots, S_{m'}$, construct a CACS instance as follows:

- Create $m' + n'$ candidates.
- Candidates $c_1, \dots, c_{m'}$ correspond to subsets; candidate c_i is approved by all voters in S_i and satisfies the constraint corresponding to the elements it covers.
- Create n' additional candidates (one per element) approved by no voters but each satisfying exactly one constraint.

- Set $\tau = d(k((2k - 3)n') + n')$.

A justified committee of size k with approval $\geq \tau$ exists if and only if a set cover of size k exists. The reduction follows the construction of [1, 3], extended with constraint satisfaction encoding. This establishes NP-hardness. $\square$

Corollary 7.2. *No polynomial-time algorithm can achieve an approximation ratio better than $\Omega(\log p)$ for CACS (under standard complexity assumptions), by reduction from Set Cover inapproximability.*

Remark 7.3. *The NP-hardness result motivates CIGAV as an approximation algorithm; its ratio of $1 + Q/k$ is meaningful because in practical settings $Q \ll k$, ensuring near-optimal solutions.*

8 Discussion and Future Work

Constraint Relaxation. When constraints cannot all be satisfied simultaneously at full strength, one can define a *priority-weighted relaxation*: assign each constraint a weight $w_t \in [0, 1]$ and relax it as $q_t \leftarrow q_t \cdot (1 - \varepsilon(1 - w_t))$. This yields a Pareto frontier between constraint satisfaction rate and voter approval, characterizing the fundamental tradeoff in constrained committee selection.

Open Problems.

1. **Tighter approximation:** Is $1 + Q/k$ tight, or can an algorithm achieve an $O(\log p)$ ratio (analogous to Set Cover)?
2. **Constraint interaction:** When constraints share satisfying candidates (e.g., rating ≥ 4 and age > 35 overlap heavily), does the approximation worsen? We conjecture a factor related to the maximum constraint overlap degree.
3. **Axiomatic study:** What axioms are compatible with hard attribute constraints? Can a constrained analogue of SJR be defined and achieved in polynomial time?
4. **Online setting:** Can CIGAV be adapted when voters and constraints arrive sequentially, with provable regret bounds?

9 Conclusion

We introduced the Constrained Approval-Based Committee Selection (CACS) problem, combining candidate-level approval ballots with externally imposed marginal attribute quota constraints. We established that this problem is NP-complete and proposed CIGAV, the first polynomial-time algorithm to jointly optimize voter representation and constraint satisfaction in this setting. The key novelty is the adaptive β -weighting mechanism that automatically shifts priority from approval maximization to constraint satisfaction as quota deadlines approach.

Our approximation analysis yields a ratio of $1 + Q/k$, showing that CIGAV is near-optimal when total quota demand is small relative to committee size—the practically relevant regime. CIGAV strictly generalizes GAV: without constraints, it reduces to GAV and inherits its Justified Representation guarantee.

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